PRACTICE EXERCISE

1. What is the value of $\log_{100} 0.1$?
(a) 1/2 (b) –1/2 (c) –2 (d) 2

2. The value of $3 \log 3 + 2 \log 2$ is
(a) log 108 (b) log 106 (c) log 109 (d) None of these

3. If $\log_a \sqrt{2} = \frac{1}{6}$, then the value of a is
(a) $(\sqrt{2})^6$ (b) $(6)^{1/2}$ (c) 3 (d) –6

4. If $\log_3 x = -2$, then the value of x is
(a) $\frac{1}{9}$ (b) $-\frac{1}{9}$ (c) $\frac{1}{8}$ (d) $-\frac{1}{8}$

5. Find the logarithm of 1728 to the base $2\sqrt{3}$.
(a) 3.124 (b) 3.1732 (c) 6 (d) 5

6. What is the value of
$(\log_{1/2} 2)(\log_{1/3} 3)(\log_{1/4} 4)\ldots(\log_{1/1000} 1000)$?
(a) 1 (b) –1 (c) 1 or –1 (d) 0

7. What is the value of
$\frac{1}{2}\log_{10} 25 - 2\log_{10} 3 + \log_{10} 18$?
(a) 2 (b) 3 (c) 1 (d) 0

8. What is the value of $[\log_{10}(5 \log_{10} 100)]^2$?
(a) 4 (b) 3 (c) 2 (d) 1

9. The value of $\log_y x \cdot \log_z y \cdot \log_x z$ is
(a) log xyz (b) xyz (c) 1 (d) 0

10. The value of $\log_3 (27 \times \sqrt[4]{9} \times \sqrt[3]{9})$ is
(a) 4 (b) $4\frac{1}{3}$ (c) $8\frac{1}{3}$ (d) $4\frac{1}{6}$

11. The value of $\log_2 [\log_2 \log_2 \log_2 (65536)]$ is
(a) 8 (b) 16 (c) 4 (d) 1

12. What is the value of $[\log_{13}(10)]/[\log_{169}(10)]$?
(a) $\frac{1}{2}$ (b) 2 (c) 1 (d) $\log_{10} 13$

13. What is the value of
$\left(\frac{1}{3}\log_{10} 125 - 2\log_{10} 4 + \log_{10} 32 + \log_{10} 1\right)$?
(a) 0 (b) $\frac{1}{5}$ (c) 1 (d) $\frac{2}{5}$

14. If $\log_r 6 = m$ and $\log_r 3 = n$, then what is $\log_r (r/2)$ equal to?
(a) $m - n + 1$ (b) $m + n - 1$ (c) $1 - m - n$ (d) $1 - m + n$

15. What is $\log_{10}\left(\frac{3}{2}\right) + \log_{10}\left(\frac{4}{3}\right) + \log_{10}\left(\frac{5}{4}\right) + \cdots$ upto 8 terms equal to?
(a) 0 (b) 1 (c) $\log_{10} 5$ (d) None of these

16. The value of $\frac{1}{\log_{xy}(xyz)} + \frac{1}{\log_{yz}(xyz)} + \frac{1}{\log_{zx}(xyz)}$ is
(a) xyz (b) 2 (c) 0 (d) 1

17. The value of
$\frac{1}{1+\log_x (yz)} + \frac{1}{1+\log_y (xz)} + \frac{1}{1+\log_z (xy)}$ is
(a) 1 (b) $\frac{1}{xy^2}$ (c) $x = yz$ (d) 0

18. If $\log_4 (x^2 + x) - \log_4 (x+1) = 2$, then the value of x is
(a) 4 (b) 8 (c) 16 (d) 1

19. If $\log_4 x + \log_2 x = 6$, then the value of x is
(a) 16 (b) 4 (c) 2 (d) 1

20. Given $\log_{10} 2 = 0.3010$, the value of $\log_{10} 5$ is
(a) 0.6990 (b) 0.6919 (c) 0.6119 (d) 0.7525

21. If $\log \frac{x}{y} + \log \frac{y}{x} = \log (x + y)$, then
(a) $x + y = 1$ (b) $x - y = 0$ (c) $x - y = 1$ (d) $x = y$

22. The characteristic in $\log 6.7482 \times 10^{-5}$ is
(a) 6 (b) – 4 (c) 5 (d) – 5

23. If $10^x = 1.73$ and $\log_{10} 1730 = 3.2380$, then x equals to
(a) 2.380 (b) 0.2380 (c) 2.2380 (d) 1.380

24. If $2^{2x+3} = 6^{x-1}$, then x equals
(a) $\frac{4\log 2 + \log 3}{\log 3 - \log 2}$ (b) $\frac{3\log 2 + 2\log 3}{\log 3 - 2\log 2}$
(c) $\frac{\log 48}{\log 7}$ (d) None of these

25. The value of $10^{\log_{10} m + 2\log_{10} n + 3\log_{10} p}$ is
(a) m^2np^3 (b) mn^2p^3 (c) m^3np^2 (d) None of these

26. Given that $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$ and $\log_{10} 7 = 0.8491$, then $\log_{10} \frac{108}{\sqrt{7}}$ is equal to
(a) 2.6123 (b) 1.6088 (c) 1.6320 (d) 2.4558

27. If a, b and c are three consecutive integers, then $\log(ac+1)$ is equal to
(a) $\log(2b)$ (b) $(\log b)^2$ (c) $2\log b$ (d) None of these

28. If $\log_r p = 2$, $\log_r q = 3$, then the value of $\log_p q$ is
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) 6

29. If $\log x^2y^2 = a$ and $\log \frac{x}{y} = b$, then $\frac{\log x}{\log y}$ is equal to
(a) $\frac{a-3b}{a+2b}$ (b) $\frac{a+3b}{a-2b}$ (c) $\frac{a+2b}{a-2b}$ (d) $\frac{a-2b}{a+3b}$

30. If $\log_{10} 5 = 0.70$, then $\log_5 10$ is equal to

(a) 1.35 (b) 1.40 (c) 1.43143 (d) 1.56

31. The value of

$\log_3\left(1+\frac{1}{3}\right)+\log_3\left(1+\frac{1}{4}\right)+\log_3\left(1+\frac{1}{5}\right)+\ldots+\log_3\left(1+\frac{1}{24}\right)$ is

(a) $-1+2\log_3 5$ (b) 2 (c) 3 (d) 4

32. If $\log(x+y)=\log x+\log y$ and $x=1.1568$, then y is equal to

(a) 7.3776 (b) $\overline{7}.3776$ (c) 5.3776 (d) 5.3116

33. If $\log_8 x+\log_4 x+\log_2 x=11$. Then, the value of x is

(a) 128 (b) 16 (c) 32 (d) 64

34. What is the value of $\log_y x^5 \log_x y^2 \log_z z^3$?

(a) 10 (b) 30 (c) 20 (d) 60

35. If $(\log_3 x)(\log x^{2x})(\log_{2x} y)=\log_x x^2$, then what is the value of y?

(a) $\frac{9}{2}$ (b) 9 (c) 18 (d) 27

36. If $\log_{10} 2, \log_{10}(2^x-1), \log_{10}(2^x+3)$ are three consecutive terms of AP, then which one of the following is correct?

I. $x=1$ II. $x=\log_2 5$

(a) Both I and II (b) Only II
(c) Only I (d) None of these

37. Consider the following statements

I. $(\log_{10} 0.1)^2+\log_{10} 10\cdot\log_{10} 100=3$

II. $\log_{10}\log_{10} 10=1$ III. $\log_{10}\sqrt{10}+\log_{10}\sqrt{10}=1$

Which of the statements given above are correct?

(a) I and III (b) II and III
(c) I and II (d) All are correct

38. If $y=(a^x)^{(a^x)\cdots\infty}$, then which one of the following is correct?

(a) $\log y = xy\log a$ (b) $\log y = x+y\log a$
(c) $\log y = y+x\log a$ (d) $\log y=(y+x)\log a$

PREVIOUS YEARS' QUESTIONS

39. What is the logarithm of 0.0001 with respect to base 10? **2012 II**

(a) 4 (b) 3 (c) – 4 (d) – 3

40. If $\log_{10} a = p$ and $\log_{10} b = q$, then what is the value of $\log_{10}(a^p b^q)$? **2012 II**

(a) p^2+q^2 (b) p^2-q^2 (c) p^2q^2 (d) $\frac{p^2}{q^2}$

41. What are the possible solutions for x of the equation $x^{\sqrt{x}}=\sqrt[n]{x^x}$, where x and n are positive integers? **2015 I**

(a) 0, n (b) 1, n (c) n, n^2 (d) 1, n^2

42. The value of $\frac{1}{5}\log_{10} 3125-4\log_{10} 2+\log_{10} 32$ is **2016 I**

(a) 0 (b) 1 (c) 2 (d) 3

ANSWERS

1	b	2	a	3	a	4	a	5	c	6	b	7	c	8	d	9	c	10	d
11	d	12	b	13	c	14	d	15	c	16	b	17	a	18	c	19	a	20	a
21	a	22	d	23	b	24	a	25	b	26	b	27	c	28	c	29	c	30	c
31	a	32	a	33	d	34	b	35	b	36	b	37	a	38	a	39	c	40	a
41	a	42	b																

HINTS AND SOLUTIONS

1. *(b)* $\log_{100} 0.1=\log_{10^2}\left(\frac{1}{10}\right)$

$=\frac{1}{2}\log_{10}\left(\frac{1}{10}\right)=\frac{1}{2}\log_{10}(10)^{-1}$

$=-\frac{1}{2}\log_{10} 10=-\frac{1}{2}$ [Rule 3]

2. *(a)* $3\log 3+2\log 2=\log 3^3+\log 2^2$

$=\log 27+\log 4$ [Rule 3]

$=\log(27\times 4)=\log 108$ [Rule 1]

3. *(a)* $\because$ $\log_a\sqrt{2}=\frac{1}{6} \Rightarrow a^{1/6}=\sqrt{2}$

$\therefore$ $a=(\sqrt{2})^6$

4. *(a)* $\log_3 x=-2 \Rightarrow x=3^{-2}=\frac{1}{3^2}=\frac{1}{9}$

5. *(c)* Let $\log_{2\sqrt{3}} 1728=x$

$\Rightarrow$ $(2\sqrt{3})^x=1728$

$\because$ $1728=2^6(\sqrt{3})^6=(2\sqrt{3})^6$

$(2\sqrt{3})^x=(2\sqrt{3})^6$

On comparing both sides, we get $x=6$

6. *(b)* $(\log_{1/2} 2)(\log_{1/3} 3)(\log_{1/4} 4)\ldots(\log_{1/1000} 1000)$

$=\left(\frac{\log 2}{\log 1/2}\right)\left(\frac{\log 3}{\log 1/3}\right)\left(\frac{\log 4}{\log 1/4}\right)\ldots\left(\frac{\log 1000}{\log 1/1000}\right)$ $\left[\because \log_b a=\frac{\log a}{\log b}\right]$

[Rule 4]

$$= \left(\frac{\log 2}{-\log 2}\right)\left(\frac{\log 3}{-\log 3}\right)\left(\frac{\log 4}{-\log 4}\right) \dots \left(\frac{\log 1000}{-\log 1000}\right)$$

$= (-1)\times(-1)\times(-1) \times \dots \times (-1)$

[$\because$ number of terms is odd]

$= -1$

7. *(c)* $\frac{1}{2}\log_{10} 25 - 2\log_{10} 3 + \log_{10} 18$

$= \log_{10} 25^{1/2} - \log_{10} 3^2 + \log_{10} 18$ [Rule 3]

$= \log_{10} 5 - \log_{10} 9 + \log_{10} 18$

$= \log_{10} \frac{5 \times 18}{9} = \log_{10} \frac{90}{9}$

$= \log_{10} 10 = 1$

8. *(d)* $[\log_{10} (5 \log_{10} 100)]^2$

$= [\log_{10} (5 \log_{10} 10^2)]^2$

$= [\log_{10} (10 \log_{10} 10)]^2$

$= [\log_{10} 10]^2$ [$\because \log_{10} 10 = 1$]

$= 1^2 = 1$

9. *(c)* $\log_y x \cdot \log_z y \cdot \log_x z$

$= \frac{\log x}{\log y} \times \frac{\log y}{\log z} \times \frac{\log z}{\log x} = 1$

$\left[\because \log_a b = \frac{\log b}{\log a}\right]$

10. *(d)* Let $\log_3 (27 \times \sqrt[4]{9} \times \sqrt[3]{9}) = x$

$\Rightarrow 3^x = 27 \times \sqrt[4]{9} \times \sqrt[3]{9}$

$\Rightarrow 3^x = 3^3 \times 3^{2/4} \times 3^{2/3}$

$3^x = 3^{25/6}$

On comparing both sides, we get

$\Rightarrow x = \frac{25}{6} = 4\frac{1}{6}$

11. *(d)* $\log_2 \log_2 \log_2 \log_2 2^{16}$

$= \log_2 \log_2 \log_2 (16)$ [$\because \log_2 2 = 1$]

$= \log_2 \log_2 \log_2 (2^4) = \log_2 \log_2 (4)$

$= \log_2 \log_2 (2^2) = \log_2 (2) = 1$

12. *(b)* $\frac{\log_{13} (10)}{\log_{169} (10)} = \frac{\log_{13} (10)}{\log_{13^2} (10)}$

$= \frac{\log_{13} 10}{\frac{1}{2}\log_{13} 10}$ $\left[\because \log_{a^b} c = \frac{1}{b}\log_a c\right]$

$= \frac{1}{1/2} = 2$

13. *(c)* We have, $\frac{1}{3}\log_{10} 125 - 2\log_{10} 4 + \log_{10} 32 + \log_{10} 1$

$= \log_{10} (125)^{1/3} - 2\log_{10} (2)^2 + \log_{10} (2)^5 + 0$ [$\because \log_{10} 1 = 0$]

$= \log_{10} 5 - 4\log_{10} 2 + 5\log_{10} 2$

$= \log_{10} 5 + \log_{10} 2 = \log_{10} 5 \times 2$

$= \log_{10} 10 = 1$ [Rule 1]

14. *(d)* Given, $\log_r 6 = m$ and $\log_r 3 = n$

$\because \log_r 6 = \log_r (2 \times 3) = \log_r 2 + \log_r 3$

$\therefore \log_r 3 + \log_r 2 = m$

$\Rightarrow n + \log_r 2 = m$

$\Rightarrow \log_r 2 = m - n$

$\therefore \log_r \left(\frac{r}{2}\right) = \log_r r - \log_r 2$ [Rule 2]

$= 1 - m + n$

15. *(c)* $\log_{10}\left(\frac{3}{2}\right) + \log_{10}\left(\frac{4}{3}\right) + \log_{10}\left(\frac{5}{4}\right) + \dots + 8\text{th term}$

$\because T_n = \log_{10}\left(\frac{n+2}{n+1}\right)$

$\Rightarrow T_8 = \log_{10}\left(\frac{10}{9}\right)$

$\therefore \log_{10}\left(\frac{3}{2}\right) + \log_{10}\left(\frac{4}{3}\right) + \log_{10}\left(\frac{5}{4}\right) + \dots + \log_{10}\left(\frac{10}{9}\right)$

$= \log_{10}\left(\frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \dots \times \frac{10}{9}\right)$ [Rule 1]

$= \log_{10}\left(\frac{10}{2}\right) = \log_{10} 5$

16. *(b)* The given expression is

$= \log_{xyz} (xy) + \log_{xyz} (yz) + \log_{xyz} (zx)$

$= \log_{xyz} (xy \cdot yz \cdot zx)$ [Rule 1]

$= \log_{xyz} (xyz)^2 = 2\log_{xyz} xyz = 2$

17. *(a)* The given expression is

$$= \frac{1}{\log_x (yz) + \log_x x} + \frac{1}{\log_y (xz) + \log_y y} + \frac{1}{\log_z (xy) + \log_z z}$$

$$= \frac{1}{\log_x xyz} + \frac{1}{\log_y xyz} + \frac{1}{\log_z xyz}$$

$= \log_{xyz} x + \log_{xyz} y + \log_{xyz} z$

$= \log_{xyz} xyz = 1$ [Rule 1]

18. *(c)* $\log_4 (x^2 + x) - \log_4 (x+1) = 2$

$\Rightarrow \log_4 \left(\frac{x^2 + x}{x+1}\right) = 2 \Rightarrow 4^2 = \frac{x^2 + x}{x+1}$

$\Rightarrow 16x + 16 = x^2 + x$

$\Rightarrow x^2 - 15x - 16 = 0$

$\therefore x = 16$ or $x = -1$ [not possible]

19. *(a)* Given, $\log_4 x + \log_2 x = 6$

$\Rightarrow \frac{\log x}{\log 4} + \frac{\log x}{\log 2} = 6$

$\Rightarrow \frac{\log x}{2\log 2} + \frac{\log x}{\log 2} = 6$

$\Rightarrow 3\log x = 12\log 2$

$\Rightarrow \log x = 4\log 2 \Rightarrow \log x = \log 2^4$

$\Rightarrow \log x = \log 16$

On comparing both sides, we get

$\therefore x = 16$

20. *(a)* $\log_{10} 5 = \log_{10} \frac{10}{2}$

$= \log_{10} 10 - \log_{10} 2 = 1 - 0.3010$

$= 0.6990$

21. *(a)* Given, $\log \frac{x}{y} + \log \frac{y}{x} = \log (x + y)$

$\Rightarrow \log \frac{x}{y} \cdot \frac{y}{x} = \log (x + y)$

$\Rightarrow \log 1 = \log (x + y)$

$\therefore x + y = 1$

22. *(d)* The characteristic in $\log 6.7482 \times 10^{-5}$ is -5.

23. *(b)* Given, $10^x = 1.73$, $x = \log_{10} 1.73$

$= \log_{10} 1730 - \log_{10} 1000$

$= \log_{10} 1730 - \log_{10} 10^3$

$= 3.2380 - 3 = 0.2380$

24. *(a)* $2^{2x+3} = 6^{x-1}$

Taking log on both sides, we get

$(2x + 3)\log 2 = (x - 1)\log 6$

$\Rightarrow 2x\log 2 + 3\log 2 = (x - 1)(\log 2 + \log 3)$

$\Rightarrow 2x\log 2 + 3\log 2 = x(\log 2 + \log 3) - \log 2 - \log 3$

$\Rightarrow x(\log 2 - \log 3) = -4\log 2 - \log 3$

$\therefore x = \frac{4\log 2 + \log 3}{\log 3 - \log 2}$

25. *(b)* $10^{\log_{10} m + 2\log_{10} n + 3\log_{10} p}$

$= 10^{\log_{10} m + \log_{10} n^2 + \log_{10} p^3}$

$\Rightarrow 10^{\log_{10} mn^2p^3} = mn^2p^3$ [$\because a^{\log_a p} = p$]

26. *(b)* $\log_{10} \frac{108}{\sqrt{7}} = \log_{10} 108 - \log_{10} \sqrt{7}$

$= \log_{10} 2^2 \times 3^3 - \log_{10} 7^{1/2}$

$= 2\log_{10} 2 + 3\log_{10} 3 - \frac{1}{2}\log_{10} 7$

$= 2 \times (0.3010) + 3(0.4771) - \frac{1}{2}(0.8491)$

$= 0.6020 + 1.4313 - 0.4245 = 1.6088$

27. *(c)* Let first integer be a.

Then, $b = a+1$ and $c = a+2$

$\therefore \quad ac + 1 = a(a+2) + 1$

$= a^2 + 2a + 1 = (a+1)^2$

$ac + 1 = b^2$

So, $\log(ac+1) = \log b^2 = 2\log b$

28. *(c)* Given, $\log_r p = 2$ and $\log_r q = 3$

By relation, $\log_p q = \dfrac{\log_r q}{\log_r p} = \dfrac{3}{2}$

29. *(c)* Given, $\log x^2 y^2 = a$

$[\because \log(mn) = \log m + \log n]$

$\therefore \quad \log x^2 + \log y^2 = a$

$\Rightarrow \; 2\log x + 2\log y = a \quad \ldots(i)$

Also, $\log x - \log y = b \quad \ldots(ii)$

On solving Eqs. (i) and (ii), we get

$\log x = \dfrac{a+2b}{4}$ and $\log y = \dfrac{a-2b}{4}$

$\therefore \; \dfrac{\log x}{\log y} = \dfrac{a+2b}{a-2b}$

30. *(c)* $\log_5 10 = \dfrac{\log_{10} 10}{\log_{10} 5} = \dfrac{1}{0.70} = 1.43143$

$[\because \log_{10} 10 = 1]$

31. *(a)* $\log_3\left(1+\dfrac{1}{3}\right) + \log_3\left(1+\dfrac{1}{4}\right)$

$+ \log_3\left(1+\dfrac{1}{5}\right) + \ldots + \log_3\left(1+\dfrac{1}{24}\right)$

$= \log_3 \dfrac{4}{3} + \log_3 \dfrac{5}{4} + \log_3 \dfrac{6}{5} + \ldots + \log_3 \dfrac{25}{24}$

$= \log_3 4 - \log_3 3 + \log_3 5 - \log_3 4 + \log_3 6 - \log_3 5 + \ldots + \log_3 25 - \log_3 24$

$= -\log_3 3 + \log_3 25 = -1 + 2\log_3 5$

32. *(a)* Given, $\log(x+y) = \log x + \log y$

$\Rightarrow \quad \log(x+y) = \log xy$

$\Rightarrow \quad x + y = xy$

$\Rightarrow y = \dfrac{x}{x-1} = \dfrac{1.1568}{1.1568-1} = \dfrac{1.1568}{0.1568}$

$= 7.37755 = 7.3776$

33. *(d)* Given, $\log_8 x + \log_4 x + \log_2 x = 11$

$\Rightarrow \quad \dfrac{\log x}{\log 8} + \dfrac{\log x}{\log 4} + \dfrac{\log x}{\log 2} = 11$

$\Rightarrow \quad \dfrac{\log x}{\log 2^3} + \dfrac{\log x}{\log 2^2} + \dfrac{\log x}{\log 2} = 11$

$\Rightarrow \quad \dfrac{\log x}{3\log 2} + \dfrac{\log x}{2\log 2} + \dfrac{\log x}{\log 2} = 11$

$\Rightarrow \quad \dfrac{11\log x}{6\log 2} = 11 \Leftrightarrow \log x = 6\log 2$

$\Rightarrow \quad x = 2^6, \;\therefore\; x = 64$

34. *(b)* $\therefore \; \log_y x^5 \log_x y^2 \log_z z^3$

$= 5\log_y x \cdot 2\log_x y \cdot 3\log_z z$

$[\because \log_a b^n = n\log_a b]$

$= 5\log_y x \cdot 2\log_x y \cdot 3 \times 1$

$= \dfrac{5\log x}{\log y} \times 2 \cdot \dfrac{\log y}{\log x} \times 3\left[\because \log_a b = \dfrac{\log b}{\log a}\right]$

$= 5 \times 2 \times 3 = 30$

35. *(b)* $(\log_3 x)(\log_x 2x)(\log_{2x} y) = \log_x x^2$

$\Rightarrow \quad \dfrac{\log x}{\log 3} \times \dfrac{\log 2x}{\log x} \times \dfrac{\log y}{\log 2x} = \dfrac{\log x^2}{\log x}$

$\left[\because \log_b a = \dfrac{\log a}{\log b}\right]$

$\Rightarrow \quad \dfrac{\log y}{\log 3} = \dfrac{2\log x}{\log x} [\because \log a^b = b\log a]$

$\Rightarrow \quad \log y = 2\log 3 \;\Rightarrow\; \log y = \log 3^2$

$[\because \log m = \log n \Rightarrow m = n]$

$\Rightarrow \quad \log y = \log 9$

$\therefore \quad y = 9$

36. *(b)* Given that, $\log_{10} 2, \log_{10}(2^x - 1)$ and $\log_{10}(2^x + 3)$ are in AP.

then,

$2\log_{10}(2^x - 1) = \log_{10} 2 + \log_{10}(2^x + 3)$

$\Rightarrow \quad \log_{10}(2^x - 1)^2 = \log_{10}(2^{x+1} + 6)$

$\Rightarrow \quad (2^x - 1)^2 = (2^{x+1} + 6)$

$\Rightarrow \quad 2^{2x} + 1 - 2.2^x = 2 \cdot 2^x + 6$

$\Rightarrow \quad 2^{2x} - 4.2^x - 5 = 0$

$\Rightarrow \; (2^x)^2 - 4(2^x) - 5 = 0$

$\Rightarrow \quad y^2 - 4y - 5 = 0$

$\therefore \quad y = 5, -1$

Hence, $\quad 2^x = -1$ or $2^x = 5$

$\therefore \quad x = \log_2(-1)$ is not possible

or $\quad x = \log_2(5)$

Then, $x = \log_2(5)$ is answer.

Hence, II is correct.

37. *(a)* I. $[\log_{10}(0.1)]^2 + \log_{10} 10 \cdot \log_{10} 100$

$= [-\log_{10} 10]^2 + \log_{10} 10 \cdot \log_{10} 10^2$

$= (-1)^2 + (1) \cdot 2\log_{10} 10$

$= +1 + (1) \cdot 2 = 1 + 2 = 3$

Hence, statement I is correct.

II. $\log_{10} \log_{10} 10 = \log_{10}(1) = 0 \neq 1$

Hence, statement II is incorrect.

III. $\log_{10}\sqrt{10} + \log_{10}\sqrt{10}$

$= \dfrac{1}{2}\log_{10} 10 + \dfrac{1}{2}\log_{10} 10$

$= \dfrac{1}{2} + \dfrac{1}{2} = 1$

Hence, statement III is correct.

38. *(a)* Given, $y = (a^x)^{(a^x)^{\cdots\infty}}$

$\therefore \quad y = (a^x)^y$

On taking log both sides, we get

$\log y = y\log a^x$

$\therefore \quad \log y = xy\log a$

39. *(c)* Let, $\log_{10} 0.0001 = x$

Then, $x = \log_{10} \dfrac{1}{(10)^4}$

$= \log_{10} 1 - \log_{10}(10)^4$

$= 0 - 4 = -4$

40. *(a)* Given, $\log_{10} a = p$ and $\log_{10} b = q$

$\log_{10}(a^p b^q) = \log_{10} a^p + \log_{10} b^q$

$\Rightarrow \log_{10}(a^p b^q) = p\log_{10} a + q\log_{10} b$

$\therefore \; \log_{10}(a^p b^q) = p \times p + q \times q$

$= p^2 + q^2$

41. *(a)* Given, $x^{\sqrt{x}} = \sqrt[n]{x^x}$, where x and n are positive integers.

On taking log both sides, we get

$\log x^{\sqrt{x}} = \log[\sqrt[n]{x^x}]$

$\Rightarrow \quad \sqrt{x}\log x = \log(x^x)^{1/n} = \log x^{x/n}$

$\Rightarrow \quad \sqrt{x}\log x = \dfrac{x}{n}\log x$

$\Rightarrow \quad \sqrt{x}\log x - \dfrac{x}{n}\log x = 0$

$\Rightarrow \quad \log x\left[\sqrt{x} - \dfrac{x}{n}\right] = 0$

$\because \quad \log x \neq 0$

$\therefore \quad \sqrt{x} - \dfrac{x}{n} = 0$

$\Rightarrow \quad \sqrt{x} = \dfrac{x}{n}$

$\therefore \quad \dfrac{x}{\sqrt{x}} = n$

On squaring both sides, we get

$\dfrac{x^2}{x} = n$

$\Rightarrow \quad x^2 - nx = 0$

$\Rightarrow \quad x(x-n) = 0$

$\therefore \quad x = 0, x = n$

Hence, the possible solution of x is 0, n.

42. *(b)* $\dfrac{1}{5}\log_{10} 3125 - 4\log_{10} 2 + \log_{10} 32$

$= \log_{10}[(5)^5]^{1/5} - \log_{10}(2)^4 + \log_{10} 32$

$= \log_{10} 5 - \log_{10} 16 + \log_{10} 32$

$= \log_{10}\left(\dfrac{5 \times 32}{16}\right) = \log_{10} = 1$

14

ALGEBRAIC OPERATIONS

Generally (3-4) questions have been asked from this chapter. Questions which are asked from this chapter are mostly based on direct identities and factor theorem.

Algebraic Expressions

A combination of constants and variables connected by the four fundamental operations +, –, × and ÷ is called an algebraic expression.

POLYNOMIALS

A polynomial is an algebraic expression consisting of variables and coefficients, having non-negative integral powers. e.g. $x^3 + 5x^2 - 1$, $3x^3 + 4x^2 y + 17$ etc.

(i) $3x^2 + 9x - 1 + 7/x$ is not a polynomial as it contains a term, namely $7/x$, having negative integral power of variable x.

(ii) $5x^2 - 7x^{7/2} + x - 1$ is not a polynomial as the term $-7x^{7/2}$ contains rational power of variable x.

Degree of a Polynomial

The exponent of the highest degree term in a polynomial, is known as degree of polynomial.

e.g. (i) $4x^3 - 9x^2 + 7x + 9$ is a polynomial with variable x of degree 3.

(ii) $4x^3 y - 3x^2 + 2xy + 1$ is a polynomial of degree 4.

Various Types of Polynomials

Polynomials are classified on the basis of degree of polynomial and number of terms which are given below.

Based on Degree of Polynomial

(i) **Constant polynomial** A polynomial of degree zero, is called constant polynomial. e.g. $f(x) = 8$.

Note The polynomial $f(x) = 0$ is called the zero polynomial. The degree of zero polynomial is not defined.